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# Marsh rice rat

The **marsh rice rat** (***Oryzomys palustris***) is a semiaquatic North American rodent in the family Cricetidae. It usually occurs in wetland habitats, such as swamps and salt marshes. It is found mostly in the eastern and southern United States, from New Jersey and Kansas south to Florida and northeasternmost Tamaulipas, Mexico; its range previously extended further west and north, where it may have been a commensal in corn-cultivating communities. Weighing about 40 to 80 g (1.4 to 2.8 oz), the marsh rice rat is a medium-sized rodent that resembles the common black and brown rat. The upperparts are generally gray-brown, but are reddish in many Florida populations. The feet show several specializations for life in the water. The skull is large and flattened, and is short at the front.

John Bachman discovered the marsh rice rat in 1816, and it was formally described in 1837. Several subspecies have been described since the 1890s, mainly from Florida, but disagreement exists over their validity. The Florida Keys population is sometimes classified as a different species, the silver rice rat (*O. argentatus*). Data from the mitochondrial cytochrome *b* gene indicate a deep divergence between populations east of Mississippi and those further west, which suggests that the western populations may be recognized as a separate species, *O. texensis*. The species is part of the genus *Oryzomys*, which also includes several others occurring further south in Mexico, Central America, and northwestern South America, some of which have previously been regarded as subspecies of the marsh rice rat. One, *O. couesi*, occurs with the marsh rice rat in Tamaulipas and southern Texas.

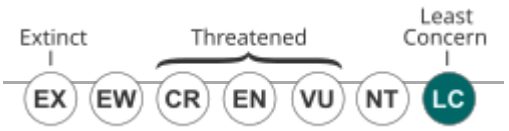
The marsh rice rat is active during the night, makes nests of sedge and grass, and occasionally builds runways. Its diverse diet includes plants, fungi, and a variety of animals. Population densities are usually below 10 per ha (4 per acre) and home ranges vary from 0.23 to 0.37 ha (0.57 to 0.91 acres), depending on sex and geography. Litters of generally three to five young are born after a pregnancy around 25 days, mainly during the summer. Newborns are helpless at birth, but are weaned after a few weeks. Several

**Marsh rice rat**

**Temporal range: Rancholabrean (300,000 years before present) – present**



**Conservation status**




Least Concern (IUCN 3.1)[1]

**Scientific classification**

| | |
|---|---|
| Kingdom: | Animalia |
| Phylum: | Chordata |
| Class: | Mammalia |
| Order: | Rodentia |
| Family: | Cricetidae |
| Subfamily: | Sigmodontinae |
| Genus: | *Oryzomys* |
| Species: | ***O. palustris*** |

**Binomial name**

***Oryzomys palustris***
(Harlan, 1837)[2]

animals prey on the marsh rice rat, including the barn owl, and it usually lives for less than a year in the wild. It is infected by many different parasites and harbors a hantavirus that also infects humans. The species is not of conservation concern, but some populations are threatened.

# Taxonomy

The marsh rice rat is classified as one of eight species in the genus *Oryzomys*, which is distributed from the eastern United States (marsh rice rat) into northwestern South America (*O. gorgasi*).[14] *Oryzomys* previously included many other species, which were reclassified in various studies culminating in contributions by Marcelo Weksler and coworkers in 2006 that removed more than 40 species from the genus.[15] All are placed in the tribe Oryzomyini ("rice rats"), a diverse assemblage of over 100 species,[16] and on higher taxonomic levels in the subfamily Sigmodontinae of the family Cricetidae, along with hundreds of other species of mainly small rodents, most of which occur in South and Central America. In the United States, the marsh rice rat is the only oryzomyine rodent except for *Oryzomys couesi* in a small area of southern Texas; the only other sigmodontines present are several species of cotton rats (*Sigmodon*) in the southern half of the country.[17]

## Early history

The marsh rice rat was discovered in 1816 in South Carolina by John Bachman.[18] Bachman intended to describe the species as *Arvicola oryzivora*, but sent a specimen to Richard Harlan and Charles Pickering at the Academy of Natural Sciences in Philadelphia to confirm its identity.[19] Another specimen, from New Jersey, was found in the academy's collection, and Harlan took it upon himself, against Pickering's wishes, to describe the new species as *Mus palustris*, proclaiming it one of the few true rats of the United States.[20] The specific name *palustris* is Latin for "marshy" and refers to the usual habitat of the species.[21]

In 1854, in *The quadrupeds of North America*, Bachman redescribed it as *Arvicola oryzivora*, considering it more closely related to the voles then placed in the genus *Arvicola*, and also recorded it from Georgia and Florida.[19] Three years later, Spencer Fullerton Baird argued that the referral of the species to *Arvicola* was erroneous and introduced a new generic name for the marsh rice rat, *Oryzomys*.[22] The name combines the Greek *oryza* "rice" and *mys* "mouse" and refers to the rat's



Current (blue) and approximate former (light blue) distribution of the marsh rice rat in the eastern United States. A small part of the distribution of *Oryzomys couesi* is also shown (red).

**Synonyms**[3]

- *Mus palustris* Harlan, 1837[2]
- *Arvicola oryzivora* Bachman, 1854[4]
- *Oryzomys palustris*: Baird, 1857[5]
- *Oryzomys palustris natator* Chapman, 1893[6]
- *?Oryzomys palustris texensis* J.A. Allen, 1894[7]
- *Oryzomys palustris coloratus* Bangs, 1898[8]
- *Oryzomys natator floridanus* Merriam, 1901[9]
- *Oryzomys fossilis* Hibbard, 1955[10]
- *Oryzomys palustris planirostris* Hamilton, 1955[11]
- *Oryzomys palustris sanibeli* Hamilton, 1955[12]
- *?Oryzomys argentatus* Spitzer and Lazell, 1978[13]

habit of eating rice.[21] At the time, *Oryzomys* was recognized either as a full genus or as a subgenus of the now-defunct genus *Hesperomys*,[22] but since the 1890s, it has been universally recognized as a genus distinct from *Hesperomys*, with the marsh rice rat (*Oryzomys palustris*) as its type species.[23]

## Species boundaries and subspecies

In the 1890s, several subspecies of the marsh rice rat were described from the United States: *O. p. natator* from Florida in 1893,[6] *O. p. texensis* from Texas in 1894,[7] and *O. p. coloratus* from elsewhere in Florida in 1898.[8] Clinton Hart Merriam recognized *O. p. natator* as a separate species in 1901 and described a subspecies of it, *O. p. floridanus*, but considered *O. p. texensis* to be nearly identical to nominate *O. p. palustris*.[24] In his 1918 revision of North American *Oryzomys*, Edward Alphonso Goldman again recognized all these as a single species, *O. palustris*. He distinguished four subspecies, which he said formed a "closely intergrading series"—*O. p. palustris* from New Jersey to southeastern Mississippi and eastern Missouri; *O. p. natator* in central Florida; *O. p. coloratus* (including *O. natator floridanus* Merriam) in southern Florida; and *O. p. texensis* from western Mississippi and southeastern Kansas to eastern Texas.[25] Two additional subspecies were described by William J. Hamilton in 1955 from southern Florida: *O. p. planirostris* from Pine Island and 2 mi (3.2 km) north of Fort Myers[11] and *O. p. sanibeli* from Sanibel Island.[12] Also in 1955, Claude W. Hibbard described a new species of *Oryzomys*, *O. fossilis*, from Pleistocene deposits in Kansas, on the basis of small differences in characters of the tooth with living marsh rice rats.[10] In 1965, Walter Dalquest demoted this species, later also found in Texas, to a subspecies, because it does not differ more from living marsh rice rats than the latter differ from each other.[26]



Distribution and subspecies of the marsh rice rat according to Goldman (1918): 1. *O. p. palustris*; 2. *O. p. natator*; 3. *O. p. coloratus*; 4. *O. p. texensis*

Merriam and Goldman had recognized that a number of Central American species, including *Oryzomys couesi* and numerous forms with more limited distributions, are related to the marsh rice rat.[27] *O. couesi* ranges north to southernmost Texas, where its distribution meets that of the marsh rice rat. In 1960, Raymond Hall argued that specimens from the contact zone were intermediate between the local forms of *O. couesi* and the marsh rice rat, and accordingly included the former in the marsh rice rat.[28] While reporting on the ecology of Texan *O. couesi* in 1979, Benson and Gehlbach noted that populations of *O. couesi* and the marsh rice rat there were in fact distinct, with the latter being smaller and less brown and more gray in color; their karyotypes were also distinct.[29] Since then, the two have generally been retained as distinct species, as supported by further research; a 1994 study even found the two to occur at some of the same places (in sympatry) in southern Texas and nearby Tamaulipas, Mexico.[30]

In 1973, rice rats were discovered at Cudjoe Key in the Florida Keys, and in 1978 Spitzer and Lazell described this population as a new species, *Oryzomys argentatus*.[13] The status of this form—either a distinct species[31] or not even subspecifically distinct from *O. palustris natator*[32]—has remained controversial since; the 2005 third edition of *Mammal Species of the World* does not recognize *O. argentatus* as a separate species, but acknowledges a need for further research.[33] A 2005 study using

microsatellite DNA found that Florida Keys rice rats exhibit low genetic variation and are significantly different from Everglades rice rats; the study concluded in favor of classifying the Keys rice rat as a "distinct vertebrate population".[34] This population probably diverged from mainland rice rats about 2000 years ago.[35]

Among the described subspecies, a 1989 morphometrical study by Humphrey and Setzer separated only two—*O. p. natator* from much of Florida (including *O. p. coloratus*, *O. p. planirostris*, *O. p. sanibeli*, and *O. p. floridanus*, as well as *O. p. argentatus*) and *O. p. palustris* from the rest of the range (including *O. p. texensis*).[36] However, Whitaker and Hamilton in their 1998 book on the *Mammals of the Eastern United States* recognized *O. p. planirostris* and *O. p. sanibeli* as separate subspecies, but merged all others into *O. p. palustris*, and placed *O. argentatus* as a separate species; their classification was based on their emphasis of overwater gaps as agents of biological diversification and a critique of shortcomings in Humphrey and Setzer's study, not on a reanalysis of the data.[37]

In 2010, Delton Hanson and colleagues published a study of the relationships among populations of *Oryzomys* on the basis of data from three genes—the mitochondrial gene cytochrome *b* (*Cytb*) and two nuclear markers, exon 1 of the interphotoreceptor retinoid-binding protein gene (*Rbp3*) and intron 2 of alcohol dehydrogenase gene 1 (*Adh1-I2*).[38] The *Cytb* data placed all marsh rice rats studied sister to a clade containing various populations of *O. couesi*; the mean genetic distance between the two groups was 11.30%. The marsh rice rats fell into two main groups, differing on average by 6.05%, one containing animals from Mississippi, southwestern Tennessee, and further west, and the other including specimens from Alabama and further east. Within the eastern group, variation was only about 0.65%, though examples of the putative subspecies *O. p. palustris*, *O. p. coloratus*, *O. p. sanibeli*, and *O. p. planirostris* were all included.[39] Data from both of the slower-evolving nuclear markers *Rbp3* and *Adh1-I2* also placed examples of *Oryzomys* in two main clades, but did not recover the western and eastern groups of the marsh rice rat as separate clades. In addition, *Adh1-I2* placed a Costa Rican population within the marsh rice rat clade and some other southern *Oryzomys* specimens closer to the marsh rice rat than to the *O. couesi* group.[40] The combined data supported the western and eastern clades within the marsh rice rat and placed the Costa Rican population marginally closer to the marsh rice rat than to *O. couesi*.[41] Using the genetic species concept, the authors suggested that the western populations of the marsh rice rat be recognized as a separate species, *O. texensis*. They recommended further research in the Mississippi–Alabama–Tennessee region, where the ranges of the two meet.[42]

## Common names

Many common names have been proposed for the marsh rice rat. Early describers used "rice meadow-mouse"[4] and "rice-field mouse"[43] and in the early 1900s, name such as "rice rat", "marsh mouse", and "swamp rice rat" came into use.[44] Some of the subspecies received their own common names, such as "Florida marsh mouse",[45] "swimming rice rat",[46] and "Central Florida rice rat" for *O. p. natator*;[47] "Bangs' marsh mouse",[45] "Cape Sable rice rat",[46] and "Everglades rice rat" for *O. p. coloratus*;[48] and "Texas rice rat" for *O. p. texensis*.[49] The species is now usually known as the "marsh rice rat",[50] although "marsh oryzomys" has also been in recent use.[51] The Florida Keys form (*O. p. argentatus*) is known as the "silver rice rat".[34]

# Description

Measurements of different populations of the marsh rice rat

| Population | *n* | Total length | Tail length | Hindfoot length |
|---|---|---|---|---|
| *O. p. palustris* (New Jersey)[52] | 4 | 242 (237–245) | 112 (109–116) | 31 (30–31.5) |
| *O. p. natator* (Florida)[53] | 10 | 281.2 (246–318) | 140.6 (122–173) | 33.1 (28–37) |
| *O. p. coloratus* (Florida)[53] | 11 | 283.0 (250–326) | 143.5 (123–171) | 33.4 (31–38) |
| *O. p. texensis* (Texas)[49] | 8 | 242 (226–279) | 120 (108–133) | 29 (28.5–30.5) |
| *O. p. planirostris* (Florida)[53] | 14 | 247.5 (226–266) | 129.6 (108–128) | 31 (29–33) |
| *O. p. sanibeli* (Florida)[53] | 11 | 257.5 (233–274) | 123.6 (111–138) | 31.0 (29–33) |
| *O. argentatus* (Florida Keys)[13] | 2 | 251, 259 | 121, 132 | 32, 32 |

Measurements are all in millimeters and are in the form "average (minimum–maximum)", except those of the Florida Keys population.

*n*=Number of specimens measured.

The marsh rice rat is a medium-sized rodent that looks much like the common black and brown rats, but has greater differences in color between the upper- and underparts.[54] The fur is thick and short.[55] The upperparts are generally gray to grayish brown, with the head a bit lighter, and are sharply delimited from the underparts, which are off-white, as are the feet. It has small cheek pouches. The ears are about the same color as the upperparts, but a patch of light hairs is in front of them. The tail is dark brown above and may be paler below.[56] The guard hairs are long and have unpigmented, silvery tips.[57] When rice rats swim, air is trapped in the fur, which increases buoyancy and reduces heat loss.[58] As in most other oryzomyines, females have eight mammae.[57]

The fore feet have four and the hind feet five digits.[59] On the fore feet, the ungual tufts (tufts of hair on the digits) are absent.[60] The hind feet are broad and have a short fifth digit. Many of the pads are reduced, as are the ungual tufts, but small interdigital webs are present.[61] The Florida Keys form, *P. o. argentatus*, has even more reduced ungual tufts.[62] Many of these traits are common adaptations to life in the water in oryzomyines.[63]

Some geographic variation in fur color occurs; western populations (*P. o. texensis*) are lighter than those from the east (nominate *P. o. palustris*), and Florida populations are generally more tawny or reddish than either, with those from southern Florida (*P. o. coloratus*) being brighter than those from the center of the state (*P. o. natator*).[65] The Florida Keys form (*P. o. argentatus*) is silvery,[66] and the two other Florida forms—*P. o. planirostris* and *P. o. sanibeli*—lack the reddish tones of mainland Florida populations and are instead grayish, resembling *P. o. planirostris*, or brownish (*P. o. sanibeli*).[67] In 1989, Humphrey and Setzer reviewed variation in color among Florida populations. They found *P. o. argentatus* to be substantially lighter and *P. o. planirostris* and *P. o. sanibeli* to be somewhat darker than mainland populations, and *P. o. argentatus* to have a less yellow fur, but found no significant differences in redness. Substantial variation within populations also was found.[68]



Marsh rice rats in much of Florida are more reddish than those elsewhere.[64]

Total length is 226 to 305 mm (8.9 to 12.0 in), tail length 108 to 156 mm (4.3 to 6.1 in), hind foot length 28 to 37 mm (1.1 to 1.5 in),[59] and body mass 40 to 80 g (1.4 to 2.8 oz), with males slightly larger than females.[66] The largest individuals occur in Florida and along the Gulf Coast east of the Mississippi River delta.[25]

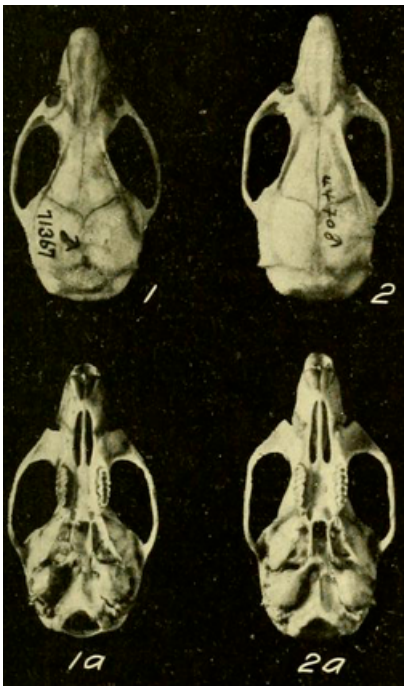
The stomach has the characteristic pattern of sigmodontines (unilocular-hemiglandular); it is not split in two chambers by an incisura angularis and the front part (antrum) is covered by a glandular epithelium.[69] The gall bladder is absent, a synapomorphy (shared-derived character) of Oryzomyini.[70] The karyotype includes 56 chromosomes and a fundamental number of 60 chromosomal arms (2n = 56, FN = 60).[71] The form of the sex chromosomes has been used to distinguish the marsh rice rat from *Oryzomys couesi*, but may be too variable among *Oryzomys* species to be useful in differentiating them.[72] X chromosome inactivation occurs in the marsh rice rat, though the animal lacks LINE-1 retrotransposons, which have been suggested as components of the inactivation process.[73] Mutants with fused or additional molars and with light fur have been recorded in laboratory colonies;[74] the abnormal molars are apparently the result of a single autosomal recessive mutation.[75] At about 50%,[76] hematocrit (the proportion of red blood cells in the blood) is high in the marsh rice rat compared to other rodents; this may be an adaptation that enables the rice rat to increase oxygen capacity while swimming under water.[77]

## Male reproductive anatomy

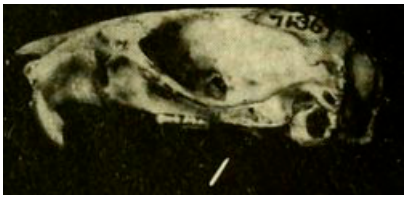
The glans penis is long and robust,[80] averaging 7.3 mm (0.29 in) long and 4.6 mm (0.18 in) broad, and the baculum (penis bone) is 6.6 mm (0.26 in) long.[81] As is characteristic of the Sigmodontinae, the marsh rice rat has a complex penis, with the distal (far) end of the baculum ending in three digits.[82] The central digit is notably larger than those at the sides.[80] The outer surface of the penis is mostly covered by small spines, but a broad band of nonspinous tissue is seen.[83] The papilla (nipple-like projection) on the dorsal (upper) side of the penis is covered with small spines, a character the marsh rice rat shares only with *Oligoryzomys* and *Oryzomys couesi* among oryzomyines examined.[84] On the urethral process, located in the crater at the end of the penis,[85] a fleshy process (the subapical lobule) is present; it is absent in all other oryzomyines with studied penes except *O. couesi* and *Holochilus brasiliensis*.[86] The baculum is deeper than it is wide.[80]

Some features of the accessory glands in the male genital region vary among oryzomyines. In the marsh rice rat, a single pair of preputial glands is present at the penis. As is usual for sigmodontines, two pairs of ventral prostate glands and a single pair of anterior and dorsal prostate glands exist. Part of the end of the vesicular gland is irregularly folded, not smooth as in most oryzomyines.[87]

## Skull



Skulls of the marsh rice rat, seen from above (top row) and below (bottom row), at the left is a South Carolina specimen of *O. p. palustris*, at the right is the holotype of *O. p. coloratus* from Cape Sable, Florida.[78]



Skull from South Carolina, seen from the left.[79]

The marsh rice rat has a large, flattened skull[59] with a short and broad rostrum.[88] The nasal and premaxillary bones extend back beyond the point where the lacrimal, frontal, and maxillary bones meet.[89] In *P. o. planirostris*, the rostrum is flatter than in mainland Florida forms, in which it is more convex,[11] and the nasals are said to be relatively longer in *P. o. argentatus*.[90] The zygomatic plate, the flattened front part of the zygomatic arch (cheekbone), is broad and develops a notch at its front end. The arches themselves are robust and contain small but distinct jugal bones.[91] The sphenopalatine foramen, an opening in the side of the skull above the molars, is large; it is much smaller in *O. couesi*.[92] The narrowest part of the region between the eyes is towards the front and the edges are lined by prominent shelves.[93] The marsh rice rat has a narrow braincase lined by prominent ridges and a narrow interparietal bone.[94] According to Goldman, Florida animals (*P. o. coloratus* and *P. o. natator*) generally have the largest and broadest skulls, and the western specimen (*P. o. texensis*) has a somewhat smaller and narrower skull than those from the east outside Florida (*P. o. palustris*).[95] In *P. o. argentatus*, the skull is also relatively narrow.[90]

The incisive foramina, openings in the front part of the palate, reach backward between the molars. The palate is long, extending substantially beyond the third molars. The back part, near the third molars, is usually perforated by prominent posterolateral palatal pits, which are recessed into fossae (depressions). The mesopterygoid fossa, the gap behind the end of the palate, is perforated by sphenopalatine vacuities, which are set far to the front.[96] The condition of the arteries in the head is highly derived.[97] The subsquamosal fenestra, an opening in the back part of the skull determined by the shape of the squamosal bone, is present.[98] The squamosal lacks a suspensory process that contacts the tegmen tympani, the roof of the tympanic cavity, a defining character of oryzomyines.[99] Some openings occur in the mastoid bone.[100]

In the mandible, the mental foramen, an opening just before the first molar, opens sidewards, not upwards as in a few other oryzomyines.[101] The upper and lower masseteric ridges, which anchor some of the chewing muscles, join at a point below the first molar and do not extend forward beyond that point.[102] The capsular process, a raising of the bone of the back of the mandible that houses the back end of the incisor, is present, but not as large as in *O. couesi*.[103]

## Teeth

The dental formula is $\frac{1.0.0.3}{1.0.0.3} \times 2 = 16$ (one upper and one lower incisor and three upper and three lower molars),[71] as usual in muroid rodents.[104] The upper incisors are well developed and strongly opisthodont, with the chewing edge located behind the vertical plane of the teeth.[105] The molars are bunodont, with the cusps higher than the connecting crests, and brachydont, low-crowned, as in most

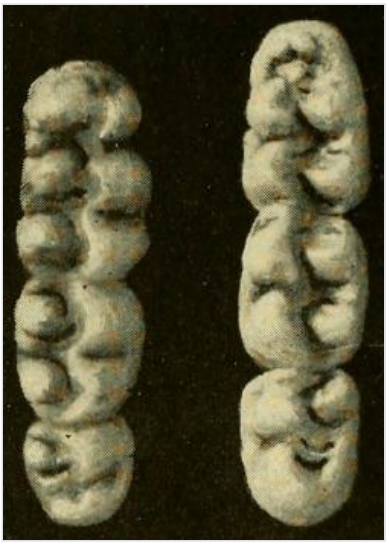


Mandible of a marsh rice rat from New Jersey, seen labially (from the outer side)[79]

other oryzomyines.[106] Many accessory crests, including the mesoloph on the upper molars and the mesolophid on the lower molars, are present, another trait the marsh rice rat shares with most but not all other oryzomyines.[107] The flexi and flexids (valleys between the cusps and crests) at the labial (outer) side of the molars are closed by cingula (ridges).[108]

The upper molars have two longitudinal rows of cusps, not three as in the black and brown rats.[109] The first and second upper molars are oval in form[110] and the flexi do not extend to the midline of the molars.[108] The anterocone, the front cusp of the upper first molar, is not divided in two by an indentation at its front (anteromedian flexus), but does display a hollow in the middle, the anteromedian fossette, which divides it into separate cuspules at the labial and lingual (inner) sides of the molar. A crest, the anteroloph, is present behind the labial cuspule, but in older animals, the cusps and the crest are united into a single structure by wear.[111] In the third upper molar, the cusps at the back are reduced and scarcely distinguishable.[112] As in most oryzomyines, the upper molars all have one root on the inner (lingual) side and two on the outer (labial) side; in addition, the first upper molar usually has another small labial root.[113]

The first lower molar is rounded at the front end and the labial and lingual conules of the anteroconid, the frontmost cusp, are barely distinct. The second lower molar is elongated and has a crest, the anterolophid, before the two cusps that form the front edge of the molar in some other oryzomyines, the protoconid and metaconid.[115] A distinct ridge (anterolabial cingulum) is at the outer front (anterolabial) edge of the molar, before the protoconid.[116] The lower third molar is about as long as the second and also has an anterolophid, albeit a less well-defined one.[117] The first lower molar has large roots at the front and back of the tooth and usually one or two smaller ones in between, at the labial and lingual side. The second and third lowers molars have either two roots, one labial and one lingual, or only one at the front, and another large root at the back.[118]


Upper (left) and lower (right) molars of a marsh rice rat from Virginia, with the front molars above[114]

## Postcranial skeleton

As usual in oryzomyines, 12 ribs are present. The first rib articulates with both the last cervical (neck) and first thoracic (chest) vertebrae, a synapomorphy of the Sigmodontinae.[119] The anapophyses, processes at the back of vertebrae, are absent from the fifth lumbar.[120] Between the second and third caudal vertebrae, hemal arches (small bones) are present with a spinous back border.[121] The entepicondylar foramen is absent, as in all members of the Sigmodontinae; if present, as in some other rodents, this foramen perforates the distal end of the humerus.[122]

### Physiology

In poor conditions, the weight of the adrenal gland may increase up to 200%,[123] and rice rats are unable to conserve water well when dehydrated,[76] and in water contaminated with oil, they swim less and their mortality increases.[124] The median amount of radiation needed to kill a marsh rice rat is 5.25 Gy and the lethal dose of potassium cyanide is 7.20 mg/kg; both values are relatively low for cricetid rodents.[125] In one study, wild rice rats in radioactively contaminated areas did not show signs of disease.[126] Exposure to more daylight and higher food availability cause increased development of the gonads in both adult and juvenile rice rats.[127] When the pineal gland is removed or melatonin is administered in male rice rats, the testes are reduced and tend to regress into the body.[128]

## Distribution and habitat



A marsh rice rat walking on mesh in Paynes Prairie, Florida

The marsh rice rat currently occurs in much of the eastern and southern United States, northeast to southern New Jersey, and south to southeastern Texas and far northeastern Tamaulipas, Mexico.[129] The northernmost records in the interior United States are in eastern Oklahoma, southeastern Kansas, southern Missouri and Illinois, and the southern half of Kentucky, but the species is absent in much of the Appalachians.[130] Fossils of the marsh rice rat are known from Rancholabrean (late Pleistocene, less than 300,000 years ago) deposits in Florida and Georgia[131] and remains referred to the extinct subspecies *O. p. fossilis* are from the Wisconsinan and Sangamonian of Texas and Illinoian and Sangamonian of Kansas.[132] In the Florida Keys, rice rats occur on most of the Lower Keys, but are absent from the Upper Keys, which are of a different geological origin and were probably never connected to the mainland.[133] The western and eastern *Cytb* clades within the marsh rice rat may represent expansions from different glacial refugia which the species was restricted to during a glacial period.[38]

Cave and archeological remains indicate that the range of the marsh rice rat has extended substantially further north and west earlier in the Holocene, into central Texas, eastern Nebraska, southwestern Iowa, central Illinois, southern Indiana, southern Ohio, West Virginia, and southwestern Pennsylvania.[134] Most northern archeological sites date from about 1000 CE and are associated with corn cultivation, but in some older cave sites the rice rat is found with the extinct giant armadillo *Dasypus bellus*, suggesting warm climatic conditions. Perhaps a warm period during the Quaternary enabled the rice rat to disperse northward and when the climate cooled, relict populations were able to survive in the north as commensals in corn-cultivating Native American communities.[135] Some subfossil animals are slightly larger than living marsh rice rats, possibly because environmental constraints were relaxed in commensal populations.[136]

In Tamaulipas and southern Texas, the ranges of the marsh rice rat and the related *Oryzomys couesi* meet;[137] in parts of Kenedy, Willacy and Cameron counties, Texas, and in far northeastern Tamaulipas, the two are sympatric (occur in the same places).[138] In experimental conditions, they fail to interbreed[139] and genetic analysis yields no evidence of gene flow or hybridization in the wild.[140]

Compared to *O. couesi*, the marsh rice rat shows less genetic variability within but more between populations in the contact zone, probably because the species is restricted to isolated populations near the coast.[141]

The marsh rice rat occurs in several habitats, ranging from coastal salt marshes to mountain streams and clearings. It is semiaquatic, spending much time in the water, and usually occurs in wetland habitats. It prefers areas where the ground is covered with grasses and sedges, which protect it from predators.[142] In southern Illinois, marsh rice rats are more likely to occur in wetlands with more herbaceous cover, visual obstruction, and nearby grasslands.[143] The species also occurs in drier uplands, which serve as sinks for young, dispersing animals and as refuges during high tide.[144] Rice rats are adept overwater dispersers; studies on islands off Virginia's Delmarva Peninsula show that they readily cross 300-m (1000 ft) channels between islands.[145]

## Behavior and ecology



The barn owl is an important predator of the marsh rice rat.

Marsh rice rats are active during the night, so are rarely seen, although they may be among the most common small mammals in part of their range. They build nests of sedge and grass, about 13 cm (5 in) large, which are placed under debris, near shrubs, in short burrows, or high in aquatic vegetation. They may also use old nests of marsh wrens (*Cistothorus palustris*), red-winged blackbirds (*Agelaius phoeniceus*), muskrats (*Ondatra zibethicus*) or round-tailed muskrats (*Neofiber alleni*). Marsh rice rats sometimes make large runways or dig burrows.[146] They are accomplished and willing swimmers, easily swimming more than 10 m (33 ft) under water,[147] and often seek safety in the water when alarmed.[148] Rice rats in the Florida Keys occasionally climb in vegetation, but never higher than 90 cm (3.0 ft).[149] Marsh rice rats are very clean and extensively groom themselves, perhaps to keep their fur water-repellent.[150] They are aggressive towards conspecifics and emit high-pitched squeaks while fighting.[74] In dense vegetation, their perceptual range (the distance from which an animal can detect a patch of suitable habitat) is less than 10 m (33 ft).[151] When released outside of their natural wetland habitat, marsh rice rats generally move either upwind or downwind (anemotaxis), perhaps to move in a straight line, which is an efficient strategy to find suitable habitat.[152]

Many animals prey on marsh rice rats. The barn owl (*Tyto alba*) is among the most important; one study found that 97.5% of vertebrate remains in barn owl pellets were marsh rice rats. Other predators include birds such as marsh hawks (*Circus cyaneus*), and barred owls (*Strix varia*); snakes such as cottonmouths (*Agkistrodon piscivorus*); alligators (*Alligator mississippiensis*); and carnivorans like raccoons (*Procyon lotor*), red foxes (*Vulpes vulpes*), American mink (*Neogale vison*), weasels (*Mustela* and *Neogale* sp.), and striped skunks (*Mephitis mephitis*).[153] Many parasites have been recorded on the marsh rice rat, including various ticks and mites, lice, and fleas among external parasites and many nematodes and digeneans, a pentastomid, and several coccidians among internal parasites (see *Parasites of the marsh rice rat*).[154]

In Florida, marsh rice rats may be eaten by some growth stage of invasive snakes such as Burmese pythons, reticulated pythons, Southern African rock pythons, Central African rock pythons, boa constrictors, yellow anacondas, Bolivian anacondas, dark-spotted anacondas, and green anacondas.[155]

Periodontitis, a bacterial disease affecting the jaws, is particularly virulent in marsh rice rats; the animal has been proposed as a model for research on the disease in humans.[156] The identity of the bacterial agent remains unknown. Vitamin E, fluoride, and iodide protect against bone loss associated with this disease in the rice rat and a high-sucrose diet increases the severity of periodontitis.[157] A case of kyphosis has been observed in a North Carolina marsh rice rat.[158]

## Population dynamics

The population density of the marsh rice rat usually does not reach 10 per ha (4 per acre).[159] The weather may influence population dynamics;[123] in the Everglades, densities may exceed 200 per ha (80 per acre) when flooding concentrates populations on small islands,[160] In the Florida Keys, population density is less than 1 per ha (0.4 per acre).[161] On Breton Island, Louisiana, perhaps an atypical habitat, home ranges in males average about 0.37 hectares (0.91 acres) and in females about 0.23 hectares (0.57 acres). A study in Florida found male home ranges to average 0.25 hectares (0.62 acres) and female 0.33 hectares (0.82 acres).[162]

Population size is usually largest during the summer and declines during winter,[160] although populations in Texas and Louisiana may be more seasonally stable.[163] Animals also often lose weight during winter.[164] Population size varies dramatically from year to year in southern Texas.[92] In coastal Mississippi, storms probably do not cause the population to decline substantially, and in Texas, inundation of its habitat did not significantly influence population density.[165] However, in Mississippi, flooding did cause a marked decline in rice rat abundance.[166]

In the northern part of its range, the species often occurs with the meadow vole (*Microtus pennsylvanicus*), but no evidence shows they compete with each other. In the south, the hispid cotton rat (*Sigmodon hispidus*) and the rice rat regularly occur together; water levels are known to influence relative abundance of these two species in Florida.[167] The cotton rat is mainly active during the day, which may help differentiate its niche from that of the rice rat.[71]

## Diet



*Spartina alterniflora* is eaten by the marsh rice rat[168]

The marsh rice rat takes both vegetable and animal food, and is more carnivorous than most small rodents are;[169] dominant food items vary seasonally. Plants eaten include species of *Spartina*, *Salicornia*, *Tripsacum*, and *Elymus*, among others; it mainly eats seeds and succulent parts.[170] It prefers *Spartina alterniflora* that has been fertilized with nitrogen and mainly eats the inner tissue of the stem, perhaps because nitrogen-fertilized plants contain much less dimethylsulfoniopropionate in their inner tissues.[168] The marsh rice rat was a major pest on rice plantations, feeding on the rice when it was newly planted. It also eats the fungus *Endogone* at times.[171]

Animals that are important to the marsh rice rat's diet include insects, fiddler crabs, and snails, but the species is known to eat many other animals, including fish, clams, and juvenile *Graptemys* and *Chrysemys* turtles. They scavenge on carcasses of muskrats, deermice, and sparrows, and may be the most important predator on eggs and young of the marsh wren.[170] Rice rats also eat eggs and young of the seaside sparrow (*Ammodramus maritimus*)[172] and are aggressive towards the sparrow, apparently leading it to avoid nesting in *Juncus*[173] in a seaside salt marsh in Florida.[172] On islands in North Carolina, rice rats consume eggs of Forster's tern (*Sterna forsteri*).[174] They have been observed preying on alligator eggs in Georgia.[175]

Laboratory studies have found that rice rats assimilate 88% to 95% of the energy in their food. They lose weight when fed on *Spartina*, fiddler crabs, or sunflower seeds alone, but a diet consisting of several of those items or of mealworms is adequate to maintain weight.[59] In an experiment, marsh rice rats did not show hoarding behavior, but wild rice rats have been observed carrying food to a nest.[74] Even when they live in uplands, they mostly eat water plants and animals, although they consume some upland plants.[144]

## Reproduction and lifecycle

Breeding occurs mostly during the summer. Some studies report that breeding ceases entirely during the winter, but winter breeding occurs as far north as Virginia, primarily because photoperiod influences their circadian rhythm which determines breeding. In both Texas and Virginia, variation in reproductive activity in females is less than in males. In the south of its range, animals may breed less when the summer is at its warmest.[176] The duration of the estrous cycle ranges from 6 to 9 days, with an average of 7.72 days. Estrus occurs again after a litter is born.[74] Copulatory behavior in the marsh rice rat is similar to that in laboratory brown rats. Before mating starts, "the male pursues the running female from behind."[177] The male then repeatedly mounts and dismounts the female; not all mounts result in an ejaculation. Penetrations only last for about 250 ms, but during mating, the penetrations[177] and the intervals between them become longer.[178] Even when a male is satiated after mating, it is able to copulate again when a new female is introduced (the Coolidge effect).[179] Partly because of resistance by the female, the frequency of ejaculation during mating is rather low in marsh rice rats as compared to laboratory rats, hamsters, and deermice.[180]

Body masses reported at different ages[123]

| Age (days) | Body mass in g (oz) |
|---|---|
| 10 | 8–17 (0.3–0.6) |
| 20 | 18–27 (0.6–1.0) |
| 40 | 27–40 (1.0–1.4) |
| 60 | 40–60 (1.4–2.1) |
| 120 | 50–80 (1.8–2.8) |

After a gestation of about 25 days, three to five young are usually born, although litter sizes vary from one to seven. Females may have up to six litters a year. Newborns weigh 3 to 4 g (about 0.11 to 0.14 oz) and are blind and almost naked. About as many males as females are born. The external ears (pinnae) soon unfold and on the first day, claws are visible and the young emit high-pitched squeaks. On the second day, they are able to crawl, and during the third to fifth days, the whiskers and eyelids develop. On the two subsequent days, the mammae and incisors become visible and the animals become more active. Between the 8th and 11th days, the eyes open, the fur develops, and the young begin to take solid food. Weaning occurs on the 11th to 20th day, according to different studies. Considerable variation is reported in body masses at different ages, perhaps because of geographic variation. Sexual activity commences when the animals are about 50 to 60 days old.[181] In the wild, rice rats usually live for less than a year;[159] one study suggested that the average lifespan is only 7 months.[182]

## Human interactions

The marsh rice rat is generally of little importance to humans, which is perhaps why it is not as well studied as some other North American rodents.[159] In 1931, Arthur Svihla noted that virtually no information had been published on the habits and life history of the marsh rice rat since the 1854 publication of Audubon and Bachman's description.[183] Writing on Everglades mammals, Thomas E. Lodge notes that although the name "rat" may associate it unpleasantly with the introduced black and brown rats, its appearance is more endearing, even cute.[184] J. S. Steward proposed the marsh rice rat as a model organism in 1951 to study certain infections to which other rodents used at the time are not susceptible.[185] The marsh rice rat is quite susceptible to periodontitis and has been used as a model system for the study of that disease.[186]

The marsh rice rat is the primary host of the Bayou virus (BAYV), the second-most common agent of hantavirus infections in the United States. About 16% of animals are infected and the virus is most prevalent in old, heavy males.[187] The virus may be transmitted among rice rats through bites inflicted during fights. It is also present in rice rat saliva and urine, and human infections may occur because of contact with these excreta.[188] Two related hantaviruses, Catacama virus and Playa de Oro virus, are known from *Oryzomys couesi* in Honduras and western Mexico, respectively.[189] An arenavirus normally associated with woodrats (*Neotoma*) has also been found in Florida marsh rice rats.[190] Antibodies against *Borrelia burgdorferi*, the bacterium that causes Lyme disease in the United States, have been found in marsh rice rats in Virginia, Maryland, North Carolina, and Tennessee.[191] Another pathogenic bacterium, *Bartonella*, is known from Georgia marsh rice rats.[192]

The 2016 IUCN Red List assesses the conservation status of the marsh rice rat as "Least Concern", because it is a common, widespread, and stable species without major threats that occurs in several protected areas.[1] The Florida Keys form is rare and in decline and is threatened by competition with the black rat, predation by domestic cats, habitat loss, and loss of genetic variation; it is considered endangered.[193] At the northern edge of its distribution, the marsh rice rat is listed as threatened in Illinois,[194] and whether it persists in Pennsylvania is unclear; it probably formerly occurred in tidal marshes on the Delaware River.[195] In Illinois, its population may have regenerated because wetlands have been developed to protect waterfowl and shorebirds and because suitable wetlands often develop in abandoned coal-mining operations.[196] A 2001 study projected that climate change would reduce the range of the marsh rice rat in Texas,[197] where it is now common, but may become threatened by habitat loss in the future.[198] A study at the Paducah Gaseous Diffusion Plant found that rice rats accumulate more polychlorinated biphenyls, but less heavy metal than white-footed mice (*Peromyscus leucopus*).[199]

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